HIERARCHICAL REINFORCEMENT LEARNING FOR TRADING

MacroHFT

A critical review and partial reproduction

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What does each architectural layer add?

We examine the trading environment, the expert pool, the aggregator and episodic memory.

01	**Execution**	Historical fills omit latency, slippage and market impact.
02	**Regime specialisation**	Trend and volatility summaries do not determine the preferred action.
03	**Q-value aggregation**	The ETH reproduction does not establish a need for an additional RL aggregator.
04	**Episodic memory**	A positive published ablation supports the module; the cause of its benefit remains unresolved.

Scope: fixed, released low-level ETH experts. The complete MacroHFT system is not reproduced end to end.

One minute, one binary position decision

The action selects a target position; the simulator trades only when that position changes.

0

Flat: no position

0.2

ETH: fixed long position

$$a_t \in \{0, 1\}, \quad P_{t+1} = a_t$$

P_t is the position before the decision. There is no limit-order execution mechanism.

State and objective

OHLCV, five order-book levels and historical context. The high level also receives rolling trend and volatility estimates.

$$J(\pi) = E_\pi[\Sigma_{t=0}^{T-1} \gamma^t r_t]$$

The reward accumulates PnL after fees. Evaluation also reports return, Sharpe ratio and maximum drawdown.

$$r_t = m[a_t(p_{t+1} - p_t) - c\, p_t|a_t - P_t|]$$

p_t: minute closing price · m: position size · c = 0.0002 (0.02%) per trade. An unchanged position incurs no fee.

> A narrow action space does not make the optimal policy trivial.

en.pdf, §§1.1–1.2, p. 1 · Equations (1)–(4); position size discussed on p. 2

The recorded price is not an execution model

Three mechanisms can separate simulated fills from the prices a live strategy obtains.

LATENCY

Information arrives, then an order travels

Data delivery, computation, the network and exchange processing all take time. Colocation only reduces part of this delay.

$$\tau = \tau_{data} + \tau_{calc} + \tau_{network} + \tau_{exchange}$$

A one-minute policy can still depend on a fleeting quote. Using the same interval's close after observing its OHLCV needs justification.

DEPTH AND SPREAD

Available liquidity is finite

A market buy pays the ask and may sweep several levels. For a 0.2 ETH order with 0.05 ETH at the best ask, the remaining 0.15 ETH fills deeper.

$$\bar{p}_{buy}(q) = (1/q)\ \Sigma_k\ p_k^{ask} q_k$$

Five-level book features do not make the closing-price reward account for this execution cost.

MARKET IMPACT

The trade can change what happens next

Orders consume liquidity and may change quotes and other participants' behaviour. Historical replay holds the market path fixed.

$$\hat{T}(s'_{mkt} \mid s, a) = \hat{T}(s'_{mkt} \mid s)$$

Impact may be small at 0.2 ETH; the simulator does not identify its size or the strategy's capital capacity.

> Simulated monetary PnL scales linearly with position size. Real execution costs need not.

Taker-only trading restricts the strategy space

The simulator excludes the choice to provide liquidity or work an order gradually.

ROUND-TRIP COMMISSION

≈ 4 bp

A purchase and a sale at c = 0.0002, before spread and slippage.

If a signal cannot cover these costs, a profitable active taker policy may not exist.

Passive execution changes the problem

A limit-order strategy can potentially earn the spread and receive maker rebates under some fee schedules.

- Orders may never fill; touching a price does not guarantee execution.
- Queue position, cancellation and order lifetime matter.
- Adverse selection can make a fill occur just before an unfavourable move.
- Multiple price levels, inventory and smaller order slices create additional decisions.

> The best policy within the taker-only set need not be the best policy among all available trading actions.

Execution can reverse the policy ranking

The review's illustrative example holds the period and capital constant.

Policy	Simulated result	Number of trades	Extra cost	Result after extra cost
A	+10 bp	20	−20 bp	−10 bp
B	**+8 bp**	**2**	**−2 bp**	**+6 bp**

Assumption: an additional loss of 1 bp of capital per trade. Simulated results already include the simulator's fees.

A backtest result is conditional

Adding an RL component can improve performance within the modelled environment without preserving the gain under live execution.

> $\hat{J}(\pi_A) > \hat{J}(\pi_B)$ does not generally imply $J_{real}(\pi_A) > J_{real}(\pi_B)$.

Capacity is a separate question

At larger order sizes, the extra cost may increase nonlinearly with liquidity consumption and market impact.

Two routes to a more informative execution test

Both ask whether the policy's advantage survives changes in latency, size and execution.

PARAMETRIC EXECUTION COSTS

$$C_t = C(s_t, q_t, \tau_t)$$

$$q_t = m|a_t - P_t|$$

Estimate spread, depth, latency, partial-fill and impact effects from order-book and trade data.

Benefit: relatively inexpensive sensitivity analysis.
Limitation: impact and participants' responses depend on the chosen model.

AGENT-BASED MARKETS

An exchange with interacting participants

ABIDES processes orders, cancellations, trades and network delays as discrete events. A policy can change both the book and future observations.

Benefit: endogenous order flow and execution.
Limitation: expensive calibration of depth, order lifetimes, liquidity recovery and reactions to trades.

Nanosecond-resolution timestamps do not by themselves make simulated behaviour realistic.

> Compare MacroHFT and simpler policies under the same execution changes, even with their trained models held fixed.

en.pdf, §1.3, p. 3 · Equation (10); Almgren–Chriss [6] and ABIDES [7]

The hierarchy changes several things at once

The comparison with EarnHFT does not isolate the contribution of volatility specialisation.

Design choice	EarnHFT	MacroHFT
Low-level trading interval	One second	One minute
High-level decision interval	One minute	One minute
Use of the expert pool	RL router selects one expert	Hyper-agent mixes Q-values
Regime specialisation	Trend-based pool	Three trend + three volatility subsets

Six subsets, not nine intersections

Train-set thresholds classify the slope of smoothed prices and return volatility. A separate dueling DDQN is trained on each subset.

Block length depends on the asset

L = 4,320 minutes for ETH, DOT and LTC;
L = 360 minutes for BTC.

The training sample, model count, timing and aggregation rule all need separate controls.

en.pdf, §2 and §2.1, p. 3 · Equation (11); MacroHFT [1], EarnHFT [2]

The same summary can imply different actions

A counterexample for linear trend without preliminary smoothing; $k > 1$.

Price history	1	2	3	4	5	Latest move
H_A	p	pk	pk^2	pk	p	Falling
H_B	p	pk^{-1}	pk^{-2}	pk^{-1}	p	Rising

Identical summaries

Both end at p and have zero linear trend. Their simple returns contain the same values in reverse order, so mean and standard deviation match.

$$\Delta^*(s) = Q^*(s, 1) - Q^*(s, 0)$$

The relevant distinction is the conditional advantage of an action.

Different conditional decisions

$$E[u_{t+1} \mid H_t] = \rho u_t, \quad \rho > 0$$

For simple returns u_t, a one-step expected-PnL choice starting flat stays flat after H_A; it enters after H_B if the expected gain covers the fee.

Conversely, different variances can yield the same action under identical conditional means and costs.

> Summary statistics lose ordering information. Smoothing or a different window does not replace a test of decision relevance.

Two trend experts largely encode directional bias

ETH test exposure: percentage of minutes long in retrospectively labelled market segments.

Expert	Bear	Neutral	Bull
slope/1	9.66	5.14	11.04
slope/2	**26.44**	**26.71**	**49.96**
slope/3	99.88	99.73	99.64

Each expert follows its own sequence of positions. The market labels are retrospective, not evidence of online regime recognition.

The pattern also appears on validation

Exposure is 5.84% for slope/1 and 99.56% for slope/3. Good performance in a preferred regime can arise from a stable flat/long preference.

Exposure explains much of the contrast

About 85–87% of the measured relative-specialisation contrast between the extreme trend agents is associated with market exposure.

This is a share of the specialisation measure, not a share of total profit.

> An almost constant action does not prove that the magnitude of the expert's Q-advantage is uninformative.

Specialisation needs controls at three levels

The gain from another model is not automatically the gain from its assigned regime.

INFORMATION

Does volatility add value?

Compare trend-only and trend-plus-volatility variants directly.

Match the number of experts: multiple seeds within each trend subset or an alternative partition can control for ensemble size.

TRAINING SAMPLE

Is separate training necessary?

Compare the same agent architecture trained on the whole market, across multiple initialisations.

One policy with regime features or balanced training examples is a direct alternative to six independent networks.

EXPERT OUTPUTS

Actions or numerical values?

Replace the extreme experts with constant flat/long policies, or remove the state dependence of their Q-advantage magnitudes.

This separates the value of position diversity from the information in continuous estimates.

The success of slope/2 outside its training category may reflect easier training, better transfer or a particular run. Its performance alone does not identify the cause.

Q-mixing changes the decision rule

The result remains a single flat/long position; capital is not split across six strategies.

Hard routing

$$i_t = g(s_t)$$

$$\pi_{hard}(s_t) = \arg\max_{a \in \{0,1\}} Q_{i_t}(s_t, a)$$

Select an expert, then execute its preferred action.

Soft Q-mixing

$$w_i(s) = \mathrm{softmax}_i\, f(s)$$

$$d_i(s) = Q_i(s, 1) - Q_i(s, 0)$$

$$\pi_{mix}(s) = 1\{\Sigma_{i=1}^{6} w_i(s) d_i(s) > 0\}$$

The weighted sum determines the position; a tie selects flat.

$$\pi_{mix}(s) \in \{\pi_1(s), \ldots, \pi_6(s)\}$$

With nonnegative weights and binary actions, the mixture agrees with at least one expert. Given the same information, a sufficiently expressive hard router can select such an expert.

> This is a statement about available actions, not a claim that hard and soft mechanisms are equally easy to learn.

Weights alone explain neither influence nor speed

Both effects depend on the magnitudes of the experts' Q-advantages.

A 10% weight can determine the action

$$0.9 \times 0.1 + 0.1 \times (-2) = -0.11$$

The mixture selects flat despite a 90%-weight expert preferring long. Influence is $w_i d_i$; a Q-advantage is an estimated return difference, not a probability of being right.

Positive rescaling preserves an individual policy but can change the mixture's action at fixed weights. Calibration therefore matters.

Two-expert exit threshold

$$d_{long} = u > 0, \quad d_{flat} = -v < 0$$

$$b_t \geq \frac{u}{u + v}$$

b_t is the flat expert's weight. Selecting the largest-weight expert switches to flat when $b_t > ½$.

20%

Soft exit when v = 4u

>50%

Largest-weight hard selection

80%

Soft exit when v = u/4

Analytical examples, not measured MacroHFT behaviour: soft mixing can switch earlier or later along the same weight path.

Rapid adaptation remains an unmeasured claim

Final returns and selected trading examples do not identify response time.

What the published evidence shows

- Figure 4 gives individual examples, without a matched EarnHFT response or a distribution of stress-event exit delays.
- Figure 5 averages expert weights over 60 minutes. It cannot resolve minute-level reaction time.

MDD, %	MacroHFT	EarnHFT
DOT	15.89	5.80
LTC	14.24	9.63

These risk metrics do not measure delay or establish uniformly better protection.

A direct test

- Use learned hard and soft controllers on the same fixed experts, inputs and trading frequency.
- Define stress events in advance; measure exit delay, event-window losses and false defensive exits.
- Account for the initial position: a policy already flat did not make a fast exit.
- Freeze weights before an event while Q-values evolve to isolate the contribution of reweighting.

> Both high-level mechanisms can update at the next minute. Mixing alone does not remove recognition delay.

en.pdf, §2.4, p. 5 · Discussion of [1], Figures 4–5 and Table 1

Hold the experts fixed; change the controller

Each strategy feeds its own previous position into the same six released ETH networks.

What is held constant

Released experts, authors' features, position size and fee. The low-level policies are not retrained.

Alternative ways to act

- Uniform Q, normalised Q-advantages and majority vote.
- A fixed expert selected on validation.
- A tabular hard router on nine trend–volatility combinations; this is not EarnHFT's RL router.
- Static and contextual supervised Ridge controllers.

Ridge predicts the next return

$$\hat{\mu}_t = b + \Sigma_{i=1}^{6} \beta_i d_i(s_t)/\sigma_i$$

σ_i is estimated on the meta-model's training segment.

$$\pi_{Ridge} = \arg\max_{a\in\{0,1\}} \{a\hat{\mu}_t - c|a - P_t|\}$$

Enter from flat if $\hat{\mu}_t > c$; exit from long if $\hat{\mu}_t \le -c$.

Train on the first validation half V_1; select regularisation on V_2. The contextual version adds trend/volatility interactions.

> This is a partial reproduction with fixed RL experts, not an end-to-end comparison of RL with a non-RL trading model.

en.pdf, §2.5, pp. 5–6 · Equations (22)–(23)

The complete aggregation comparison

ETH test, June–October 2023. Selection and tuning use validation data.

Method	MTM return, %	Author-style return, %	Sharpe	MDD, %	Exposure, %
Uniform Q	5.72	5.14	0.773	12.92	29.64
Normalized Q	−4.04	−3.53	−0.333	18.07	34.31
Majority	9.08	9.31	1.128	9.65	35.38
Tabular hard router	10.04	9.94	0.915	18.17	82.93
Ridge-Static, $V_1 \rightarrow V_2$	13.36	13.22	1.414	12.96	46.80
Ridge-Contextual, $V_1 \rightarrow V_2$	6.88	7.22	0.992	9.37	24.67
Fixed expert: selected by validation return	**23.28**	**23.87**	**2.424**	**9.54**	**32.09**
Fixed expert: selected by validation Sharpe	−22.47	−18.74	−3.116	25.28	22.80

MTM return, Sharpe and MDD use a common initial capital. Author-style return divides by **required_money**. Exposure is the percentage of minutes spent long.

A fixed expert is an essential aggregation baseline

The strongest result in this reproduction depends on how the expert is selected.

Selection by validation return

slope/2 yields 23.28% MTM return versus 9.08% for Majority, with slightly lower exposure (32.09% vs 35.38%) and similar MDD (9.54% vs 9.65%).

More time long does not explain the gap.

Selection by validation Sharpe

The original procedure selects vol/2 and gives a negative test return. This limits the claim to the policies and experiment studied.

Normalisation worsens results; the hard router raises drawdown and exposure; contextual Ridge trails static Ridge in return.

13.22%

Static Ridge · our reproduction

Author-style return

12.73%

MacroHFT without memory

Published return

39.28%

Full MacroHFT

Published return

> Numerical proximity motivates a controlled comparison. It does not establish equivalence to the authors' trained model.

en.pdf, §2.5, p. 6 · Table 2 and the published no-memory/full results in [1]

Memory supplies an additional training target

The motivation combines unstable rewards in similar states and insufficient use of rare fluctuations.

A table separate from the network

$$M_{mem} = (K, E, V)$$

K: retrieval keys
E: state–action entries
V: stored action-value estimates

The encoder creates $h_i = \psi_{\theta_j}(s_i)$. Capacity is finite; FIFO replaces the oldest entry.

Episodic Memory DQN and Neural Episodic Control already explored related memory-based learning ideas [8, 9].

What the stored value means

$$v_i = r_i + \gamma \max_{a'} Q_{\theta_j}(s'_i, a')$$

The future-value term is zero at a terminal transition.

Only r_i is directly observed. The remaining value is the network's own prediction: a stored one-step bootstrapped target.

It is not the realised profit from the rest of a trade or a completed episode, nor a permanent archive of dangerous events.

> At test time there is no memory lookup. Memory affects trading through parameters learned during training.

en.pdf, §§2.6.1–2.6.2, p. 6 · Equations (24)–(26); implementation [12]

Retrieval averages nearby stored estimates

The review follows the published implementation where it differs from the paper's formula.

Retrieve, filter and average

Find k nearest keys in Euclidean distance, then use only neighbours with the requested action.

$$\kappa(h, h_i) = \frac{1}{\|h - h_i\|^2 + \varepsilon}$$

$$\omega_i \propto \kappa(h, h_i)1\{a_i = a\}, \quad \Sigma_i \omega_i = 1$$

$$Q_M(s, a) = \Sigma_{i \in N_k(h)}\, \omega_i(s, a) v_i$$

The paper's equation (7) uses distance rather than its inverse. Relative neighbour weights therefore differ.

Train against the auxiliary target

$$L(\theta) = L_0(\theta) + \lambda_M E[(Q_\theta - Q_M)^2]$$

L_0 contains the other original terms, including TD error and an auxiliary dynamic-programming target.

Q_M is computed during transition collection and stored in the ordinary replay buffer.

Before memory fills, or if no retrieved neighbour has the same action, the code falls back to the ordinary one-step estimate.

> Faster learning from transitions and faster exits in a new market are different properties.

en.pdf, §2.6.2, pp. 6–7 · Equations (27)–(30); memory.py and high_level.py [12]

The memory ablation is a strong positive result

The complete table from the review. We did not retrain the hyper-agent.

	Without memory			Full MacroHFT		
Market	**TR, %**	**ASR**	**MDD, %**	**TR, %**	**ASR**	**MDD, %**
BTC	2.03	0.49	6.49	3.03	0.61	5.41
ETH	**12.73**	**1.26**	**20.87**	**39.28**	**3.89**	**9.67**
DOT	2.41	0.34	27.23	13.79	0.97	15.89
LTC	−8.66	−0.58	22.03	18.16	1.50	14.24

TR: return · ASR: annualised Sharpe ratio · MDD: maximum drawdown.

All four assets improve

Return and Sharpe rise; MDD falls. Return improves by 26.55 percentage points on ETH and 26.82 on LTC.

Robustness remains unmeasured

No distribution across repeated initialisations is reported here; the assets' test periods overlap substantially in calendar time.

en.pdf, Table 3, p. 8 · Reproduced from Zong et al. [1], Table 3; published results

Rare-event learning is one possible explanation

Storage and retrieval in MacroHFT do not directly prioritise the size or rarity of losses.

What this memory selects

- FIFO favours recent observations.
- Retrieval favours nearby encoded states.
- A rare event can be evicted or seldom retrieved.

Its information may still survive in network parameters and the ordinary replay buffer.

Controls for the sampling hypothesis

- **Ordinary replay:** preserves the frequency of observed transitions in expectation.
- **PER:** resamples high-TD-error transitions and uses importance-sampling corrections. Large errors need not mean financial disasters.
- **Event-based sampling:** deliberately represents pre-defined sharp moves or losses in training batches.

Increasing the weight of crisis observations without correction changes the training distribution. If that is intended to induce caution, it is part of the objective, not merely a remedy for too few examples.

Figure 6 illustrates a late ETH exit without memory; it does not isolate how much of the gain comes from avoiding rare losses.

> Measure which events enter training, how often they are retrieved and whether losses fall in new stress episodes.

en.pdf, §§2.6.3–2.6.4, pp. 7–8 · Rare-transition hypothesis; Prioritized Experience Replay [10]

A second target can also stabilise Q-values

This algebra describes two fixed quadratic targets, not the solution of full network training.

$$\ell(q) = (q - y)^2 + \lambda_M (q - z)^2 \quad \Rightarrow \quad q_{\text{eff}} = \frac{y + \lambda_M z}{1 + \lambda_M}$$

y: fixed TD target · $z = Q_M(s, a)$: fixed memory estimate · $q = Q_\theta(s, a)$.

When smoothing may help

Nearby entries can contribute information about the same conditional value, reduce target variance and alter optimisation even in ordinary states.

Auxiliary Q-targets, value averaging and regularisation are natural controls for this explanation.

What averaging cannot guarantee

Similar embeddings may merge states with different action consequences, particularly near regime boundaries.

Noise, estimation error and missing state information are different causes of reward dispersion. Averaging cannot recover a missing feature.

Bootstrapped values may smooth errors or propagate a systematic error; they are not independent verification.

> Better agreement with a model's own target does not establish better prediction of future rewards.

Remembering risk does not specify a risk objective

Even perfect recall can leave a risky positive-expected-reward strategy preferable.

What MacroHFT optimises

Its auxiliary memory loss aligns Q-values. It does not impose an explicit tail-loss constraint.

The lower drawdown is an empirical training outcome, not a direct consequence of an optimised risk criterion.

ALTERNATIVE OBJECTIVE FROM THE REVIEW

$$J_{risk}(\pi) = E[G_\pi] - \lambda_R CVaR_\alpha(-G_\pi)$$

G_π is total reward; $\lambda_R \geq 0$ controls the penalty. At $\alpha = 0.95$, CVaR is the mean loss in the worst 5% of outcomes.

This defines a different optimisation problem. It is a proposed comparison, not a description of a hidden memory effect.

Compare the trade-off

Evaluate tail losses, drawdowns and foregone return.

Report exposure

Less time in the market is not equivalent to better recognition of dangerous states.

> Memory can improve reward estimation while a risk-sensitive objective changes preferences; the mechanisms can complement each other.

en.pdf, §2.6.4, p. 8 · Equation (33); risk-sensitive RL [11]

Separate the mechanism from the performance gain

The review proposes comparisons that target distinct explanations.

Hypothesis	Matched comparison	What to measure
Rare transitions	Ordinary replay, PER, event-based sampling, memory	Crisis participation and retrieval; new stress-event losses
Useful neighbours	Nearest states vs random same-action, similar-age entries; uniform neighbour weights	Retrieval contribution; held-out Q-value quality
Smoothing	Same auxiliary loss with alternative Q_M; averaging and regularisation	Prediction quality beyond agreement with own targets
Risk preferences	Memory vs an explicit risk objective	Tail loss, drawdown, foregone return and exposure

Keep training data, low-level experts, hyper-agent architecture, tuning budgets and update budgets comparable. Report training time and retrieval cost separately.

> Repeat initialisations and use non-overlapping test periods before attributing a robust gain to a particular mechanism.

Promising results; several explanations still need tests

A successful combination of components does not by itself identify why it works.

Execution bounds the claim

Latency, liquidity and capital scale can change both returns and rankings. Transfer to realistic execution remains an empirical question.

The hierarchy needs separate controls

Volatility information, data partitioning and model count need separate controls. A training label does not ensure detection of a regime change.

Aggregation needs stronger baselines

One validation-selected expert earns more than every tested aggregator, but selection by Sharpe fails. The incremental value of RL aggregation remains unresolved.

Memory has direct positive evidence

The published ablation improves all three metrics on all four assets. Rare-event learning, smoothing and risk explanations remain to be distinguished.

> Whether RL adds value over simpler trading models remains open. Our reproduction uses fixed RL experts and does not test the complete system end to end.

Sources cited in the review

The presentation's substantive source is the accompanying en.pdf.

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